In-frame deletion of lasR in LB014 via pEXG2 allelic exchange

Primer design • PCR / Assembly protocol • 2026-05-07

1 Konstrukt-Überblick

Zielgen	lasR in LB014 (720 nt, 239 aa)
Strategie	In-frame Deletion, Scar 17 aa (15 N + 2 C)
Vektor	pEXG2 (5084 bp), HindIII-linearisiert
Methode	In-Fusion 3-Fragment (Vektor + UP + DN)
Counterselektion	sacB / GmR (pEXG2)
Finales Plasmid	pEXG2-ΔlasR_LB014 • 6113 bp

2 Primer

Primer	Sequenz 5'→ 3' (Tail fett, Body normal)	nt	Tm_{body}	GC
P1_lasR_UP_Fwd	**CATAAATGTAAAGCA**CTGTTCTGCGTCGCCGAAC	34	61.6 °C	63 %
P2_lasR_UP_Rev	**GCAAGATCAGAGAGT**TCCACTTGAGCGTTCCAGCTC	36	61.7 °C	57 %
P3_lasR_DN_Fwd	**GAACGCTCAAGTGGA**ACTCTCTGATCTTGCCTCTCAGGTC	40	61.5 °C	52 %
P4_lasR_DN_Rev	**CGACCTGCAGAAGCT**CCAGCCTTTGCGCTCCTTG	34	61.6 °C	63 %

3 PCR-Bedingungen (B7 High Fidelity)

Tm_{body} Mittel	61.6 °C (Spread 0.16 °C)
Annealing Ta	62 °C (= Tm_{body}; Gradient ± 5 °C empfohlen falls 1. Versuch fehlschlägt)
Elongation	60 s/kb • UP 588 bp → 35 s • DN 501 bp → 30 s

4 Konstrukt-Zusammenfassung

UP-Amplicon 588 bp • DN-Amplicon 501 bp • Insert 1059 bp • Final 6113 bp

Scar-Sequenz (15 N + 2 C aa + Stop):

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ATG GCC TTG GTT GAC GGT TTT CTT GAG CTG GAA CGC TCA AGT GGA | ACT CTC TGA
M A L V D G F L E L E R S S G | T L *
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Off-Target Scan: PASS • HindIII-Sites in finalem Plasmid: 0 (erwartet 0)